

GRUM GUARD

Secure Cloud File Storage Application

OVERVIEW

GRUM Guard is an application that can securely store and retrieve your personal files using the Cloud. All encryption is done locally using a private key which limits the opportunities an adversary has to intercept and interpret traffic.

PROBLEM

As technology advances, the use of the Cloud for file storage is becoming an increasingly popular trend. However, this gives adversaries many new opportunities to obtain your personal information.

SOLUTION

Our solution to this problem is to offer a method in which files can be securely stored, searched, and retrieved, without revealing any of the data in question.



Store Files



Search Files



Privacy

TECHNOLOGIES USED



PROCESS

1. Encrypt files and upload to Amazon S3 file storage

- Files encrypted with AES in CTR mode
- Use of multithreading to improve encryption speed

2. Map hashed keywords to encrypted file identifiers locally

- Keywords are parsed from file and hashed with SHA-256
- File identifier is encrypted with nonce to make them unique
- A map of keyword to identifier is created from all uploaded files

3. Index keywords into database

- All communications done over HTTPS for extra security
- Server indexes encrypted data without learning information

4. Query for files containing specific keywords. Obtain file IDs

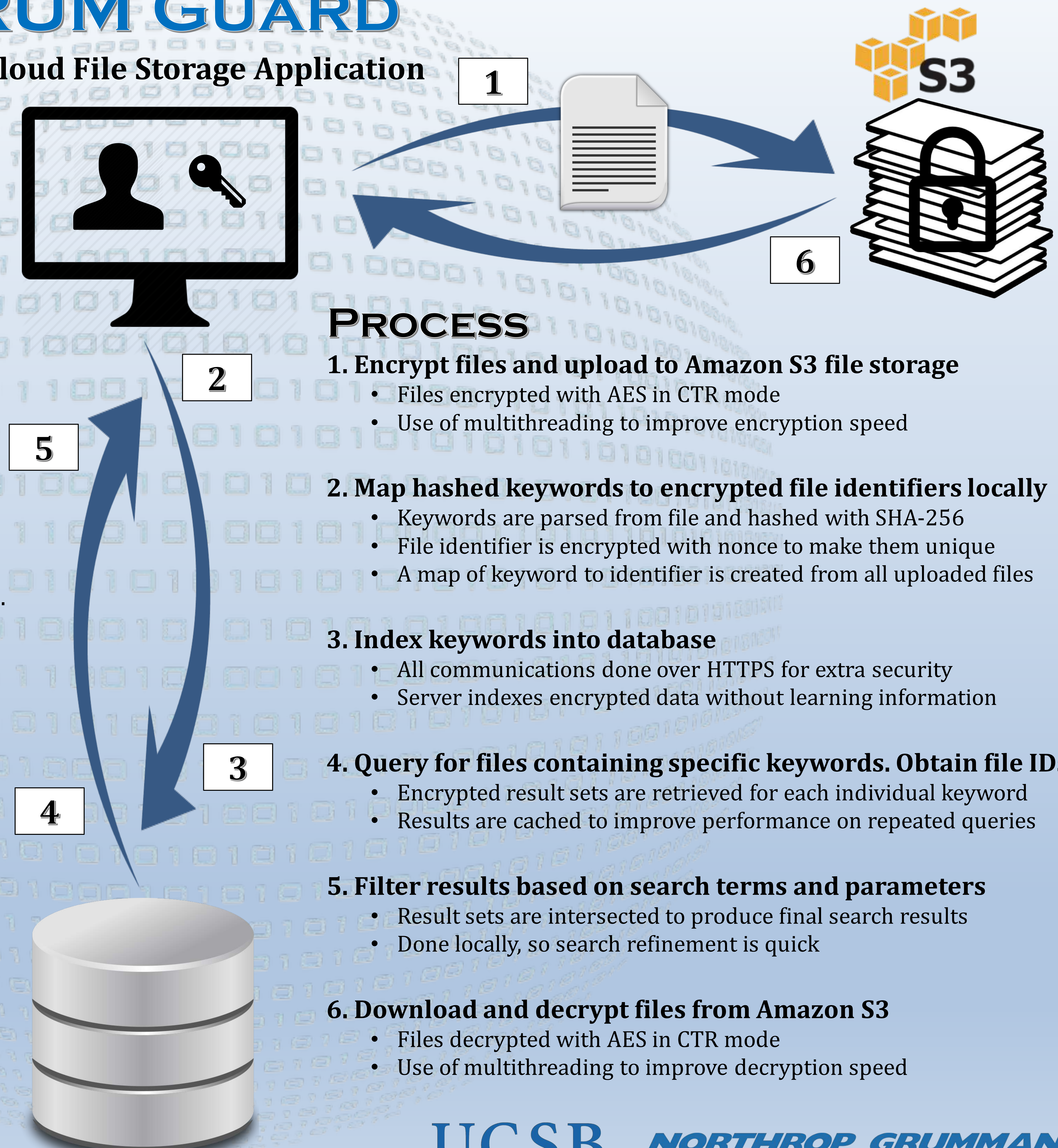
- Encrypted result sets are retrieved for each individual keyword
- Results are cached to improve performance on repeated queries

5. Filter results based on search terms and parameters

- Result sets are intersected to produce final search results
- Done locally, so search refinement is quick

6. Download and decrypt files from Amazon S3

- Files decrypted with AES in CTR mode
- Use of multithreading to improve decryption speed



Hopefully Unhackable

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